

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

1. Let  $f(x) = x^2 - 3$ . Evaluate each expression.

(a)  $f(0) =$  \_\_\_\_\_

(b)  $f(-2) =$  \_\_\_\_\_

(c)  $f(x + 1) =$  \_\_\_\_\_

2. The table below gives values of a function  $f$ .

|        |    |    |   |   |   |
|--------|----|----|---|---|---|
| $x$    | -2 | -1 | 0 | 1 | 2 |
| $f(x)$ | 5  | 2  | 1 | 2 | 5 |

State the domain and range of  $f$  using the values in the table.

Domain: \_\_\_\_\_ Range: \_\_\_\_\_

3. The graph of  $p(x)$  is a polynomial with the following features: it passes through  $(-2, 0)$ ,  $(1, 0)$ , and  $(0, -2)$ , and its end behavior is  $p(x) \rightarrow +\infty$  as  $x \rightarrow +\infty$  and  $p(x) \rightarrow -\infty$  as  $x \rightarrow -\infty$ .

Circle the best description of the degree and leading coefficient of  $p(x)$ .

- A.** Even degree, positive leading coefficient      **B.** Odd degree, positive leading coefficient  
**C.** Even degree, negative leading coefficient      **D.** Odd degree, negative leading coefficient